WASP-33b

R-0288

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_WASP-33_180125` · created 2026-07-17T17:19:58+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458143.641 +/- 0.017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.039
Transit depth (Rp/Rs)^2	1.6 +/- 0.98 [%]
Orbital Inclination (inc)	87.91 +/- 2.36
Airmass coefficient 1 (a1)	0.844 +/- 0.021
Airmass coefficient 2 (a2)	0.112 +/- 0.016
Scatter in the residuals of the lightcurve fit is	2.42 %
Best Comparison Star	#3 - [280, 355]
Optimal Aperture	10.16
Optimal Annulus	14.4
Transit Duration (day)	0.112 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.126 ± 0.039 vs 0.1066 (deviation 0.4σ)
Duration fit vs published	0.112 d vs 0.1196 d (deviation 0.51σ)
Mid-time O–C	-30.7 min
Depth SNR	2.6
Residual scatter	2.42 %

Light curve

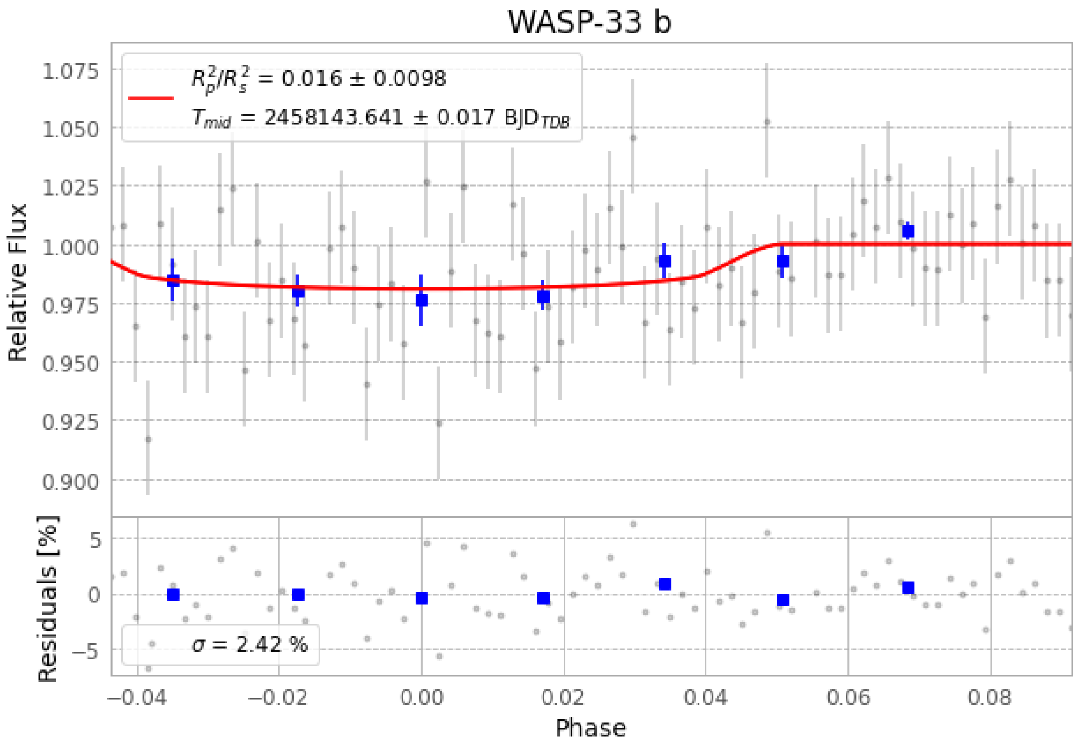

Figure 1 — FinalLightCurve_WASP-33 b_2018-01-24.png (SHA-256 50b032ed02e3c26d…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 302], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 b5b1b253f3d8ba0c…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit faa81f0`

Data provenance

80 FITS frames (53 MB), `WASP-33180125020312.FITS` → `WASP-33180125060012.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-33-180125.log (9dd44c6f0356…), FinalLightCurve_WASP-33 b_2018-01-24.png (50b032ed02e3…), FinalLightCurve_WASP-33 b_2018-01-24.csv (8b9e3fafe0a6…)

Compute resources

Wall time 155.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)